

1 equations

ALL IN THIS 1-2:

$$\frac{A_2}{B_2} = 8\pi\bar{h}\frac{\mu^3}{\lambda^3};$$

$$g_1B_{12} = g_2B_{21}$$

$$-dn_2(\nu) = A_{21}n_2f(\nu)d\nu;$$

$$-dn_2 = B_{21}n_2f(\nu)\rho_\nu d\nu$$

$$f(\nu) = \frac{\frac{\Delta\nu}{2\pi}}{(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2};$$

$$\Delta\nu = \frac{1}{2\pi\tau};$$

$$\text{GAP} = \sqrt{1 + \frac{I}{I_s}\Delta\nu};$$

$$\text{Stable } \textbf{Condition}: 0 < (1 - \frac{L}{R_1})(1 - \frac{L}{R_2}) < 1;$$

$$\Delta n = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}};$$

$$\Delta n = n_2 - \frac{g_2}{g_1}n_1;$$

$$G = \Delta n B_{21} f(\nu) h \nu \frac{m u}{c};$$

$$\Delta n = R_2 \tau_2 - (R_1 + R_2) \tau_1;$$

$$G = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]G^0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{G^0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}};$$

$$G^0(\nu) = \Delta n_0 B_{21} \frac{\mu}{c} f(\nu) h \nu_0$$

$$\Delta n(\nu) = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0 f_D(\nu)}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0 f_D(\nu)}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}};$$

$$\text{wide of hole } \nu \pm \sqrt{1 + \frac{I}{I_s} \frac{\Delta\nu}{2}}$$

$$G_D(\nu_1) = \frac{G_d^0(\nu_1)}{\sqrt{1+\frac{I}{I_s}}}$$

$$a_{all} = a_{in} - \frac{1}{2L} \ln r_1 r_2 = -\frac{1}{2L} \ln(1 - (a_1 + t_1))$$

$$\textbf{Condition}: 2\delta\Phi = 2q\pi, q = 1, 2, 3...$$

$$\Delta\nu_q = \frac{c}{2\mu L}$$

$$2L = q\lambda$$

$$\text{square} \Delta\Phi_{mn} = (m+n+1)\frac{\pi}{2}$$

$$\nu_{nmq} = \frac{c}{2\mu L}(q + \frac{1}{2}(m+n+1))$$

$$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \Delta\nu_n$$

$$\text{round} \Delta\Phi_{mn} = (m+2n+1)\frac{\pi}{2}$$

$$\nu_{nmq} = \frac{c}{2\mu L}(q + \frac{1}{2}(m+2n+1))$$

$$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \frac{1}{2}\Delta\nu_n$$

$$\textbf{Gauss:}$$

$$\omega(z) = \omega_0 \sqrt{1 + (\frac{f}{z})^2}$$

$$\omega_0 = \sqrt{\frac{\lambda f}{\pi}}$$

$$R(z) = z[1 + (\frac{f}{z})^2]$$

$$L = |z_1| + |z_2|$$

$$R_1 = R_2 = 2f(\text{S.C chamber}) \text{ not suitable for normal shpere.}$$

$$f = \frac{\pi\omega_0^2}{\lambda}$$

$$2\theta = \frac{2\lambda}{\pi\omega_0}$$

$$\begin{aligned}
q &= z + if \\
2\theta_{mn} &= \sqrt{2m+1}\sqrt{2n+1}2\theta_0 \\
V_{00}^0 &= \frac{L^2\lambda}{2} \\
V_{mn}^0 &= \sqrt{2m+1}\sqrt{2n+1}V_{00}^0 \\
&\text{HOMOS} \\
G &= \frac{G^0}{1+2\frac{I}{I_s}} = G_{Thres} \\
a_{all} &= \frac{a_1+t_1}{2L} \\
I &= \frac{I_s}{2}t_1\left(\frac{2LG^0}{a_1+t_1} - 1\right) \\
P &= IS \\
&\text{non-HOMOS} \\
\text{non-center franquency :} G(\nu) &= \frac{G_D^0(\nu)}{sqr{t1+\frac{I}{I_s}}} = G_{Thres} \\
\text{center franquency :} G(\nu) &= \frac{G_D^0(\nu)}{sqr{t1+2\frac{I}{I_s}}} = G_{Thres}
\end{aligned}$$

2 homeworks

all:

chapter1: 2,3,4,11,12,13 P28

chapter2: 1,2,3,7,8,10,11,12,13 P48

chapter3: 1,2,4,6,7,9,10 P91

chapter4: 1,2,3,4,5,8,9 P119